

Total Marks: 60
Time Allowed: 2 Hours

Section A – Short Questions (5 Marks Each)

(Attempt all questions)

1. Your company is developing a new **mobile app**. The project is currently in the **execution phase**, but the client is frequently requesting changes. As a project manager, how will you handle this situation?
 2. Your project is facing budget constraints, and the client has also reduced the deadline. How does the **Scope Triangle** apply in this case, and what trade-offs must be considered?
 3. A software company is developing an **online banking system**. How would you create a **Work Breakdown Structure (WBS)** for this project?
 4. Your project is running smoothly, but suddenly, key team members resign. What risk management strategies can be applied to mitigate the impact?
 5. A project is halfway completed, but costs have exceeded the budget. How can you use **EVA (Earned Value Analysis)** to determine the project's financial status?
 6. A project team is using different estimation techniques to determine the duration of activities. What are the five common methods used for estimating activity duration?
-

Section B – Long Questions (10 Marks Each)

(Attempt any **three** questions)

7. Your company is launching a **hospital management system**. Midway through the project, the client **changes the requirements** and requests additional features. Discuss how you will handle this situation, considering **Triple Constraints (Scope, Time, Cost)** and **Change Management techniques**.
8. Your software development project consists of the following tasks:

Task	Predecessor	Duration (Days)
A	-	3
B	A	5
C	A	2
D	B, C	4
E	C	6
F	D, E	7

- Draw the **Project Network Diagram**.
 - Identify the **Critical Path**.
 - Calculate the **Total Project Duration**.
9. A software development project has the following details at **Month 4**:
 - **Planned Value (PV)**: \$50,000

- **Earned Value (EV):** \$40,000
- **Actual Cost (AC):** \$45,000

Calculate and interpret the following:

- Schedule Variance (SV)
 - Cost Variance (CV)
 - Schedule Performance Index (SPI)
 - Cost Performance Index (CPI)
 - Explain whether the project is ahead/behind schedule and over/under budget.
10. Your company is planning to bid for a **smart home automation system** project.
- What key components should be included in the **Project Proposal**?
 - Explain different **team structures** used in project staffing, such as **Controlled Decentralized, Democratic Decentralized, and Controlled Centralized**.

